

Machine Learning and IoT Based Traffic Management System

Prioritizing Emergency Vehicles and Reducing Congestion

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Outline

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Traffic Crisis in Dhaka

- **Severe Congestion:** Average vehicle speed drops to 4.8 km/h
- **Economic Impact:** \$3.8 billion annual cost
- **Emergency Delays:** 56% of emergency responses delayed
- **Infrastructure Limits:** Roads designed for smaller population
- **Mixed Traffic:** Complex vehicle types competing for space



Figure: Shahbag Traffic Congestion

Need for Intelligent Traffic Management

Current Traffic Management Challenges

Technical Challenges:

- Fixed timing signals
- No real-time adaptation
- Manual traffic control
- Limited sensor integration
- Poor emergency prioritization

Impact on Society:

- 22 million population affected
- Daily productivity losses
- Increased fuel consumption
- Emergency service delays



Figure: Traffic Congestion at Major Intersections

Traditional systems cannot handle dynamic traffic patterns

Research Motivation

1. Urbanization Crisis

- Population growth exceeds infrastructure development
- Vehicle ownership increasing exponentially

2. Technology Opportunity

- IoT and ML technologies are now mature
- Real-time processing capabilities available

3. Life-Critical Need

- Emergency services facing critical delays
- Traffic accidents on the rise

4. Economic Necessity

- Massive economic losses due to traffic
- Cost-effective solutions needed



Figure: Traffic Scenario in Narayanganj

Key Objectives

1. Real-time Vehicle Detection

- YOLOv11 based object detection
- Classify vehicles, pedestrians, emergency vehicles

2. Emergency Vehicle Prioritization

- Automatic detection and signal adjustment
- Reduce emergency response times

3. Adaptive Traffic Control

- Dynamic signal timing based on traffic density
- Prevent lane starvation

4. IoT Integration

- Arduino, Raspberry Pi, NodeMCU
- Connect ML algorithms with physical infrastructure

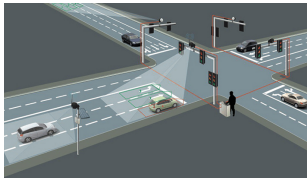
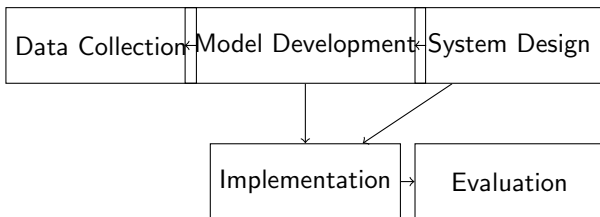


Figure: CCTV Infrastructure

Research Methodology Overview



5-Phase Systematic Approach:

- Phase 1: Comprehensive traffic data collection from 7 strategic locations
- Phase 2: YOLOv11 model training and optimization
- Phase 3: IoT-based system architecture design
- Phase 4: Real-world pilot deployment
- Phase 5: Performance evaluation and analysis

Data Collection Strategy

Strategic Locations:

1. Shahbag (Academic area)
2. Polton (Residential/Commercial)
3. Motijheel (Business district)
4. Science Lab (Transit hub)
5. Panthapath (Major arterial)
6. Bijoy Sarani (North-South corridor)
7. Gulistan (Dense urban area)

Data Characteristics:

- 6-month collection period
- Multiple time periods
- Various weather conditions
- High-resolution imagery
- Diverse traffic scenarios

Quality Assurance:

- Multi-level annotation review
- Expert domain validation
- Consistency checks
- Data augmentation techniques

System Architecture

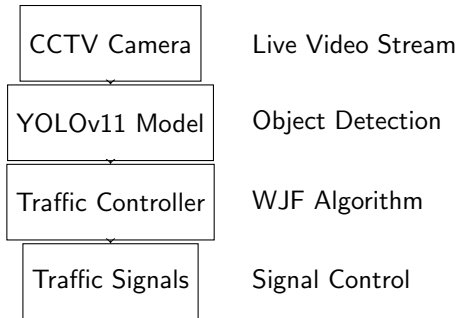


Figure: System Architecture Diagram

YOLOv11 Model Architecture

Why YOLOv11?

- Real-time processing capability
- Superior accuracy vs previous versions
- Efficient memory usage
- Multi-scale object detection
- Robust performance in challenging conditions

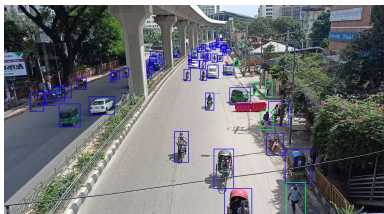


Figure: YOLOv11 Detection Example

Model Specifications:

- Input: 640x640 pixels
- Backbone: CSPDarkNet53
- Neck: PANet
- Head: YOLOv11 Detection Head

Training Details:

Parameter	Value
Model	YOLOv11m
Epochs	256
Batch Size	16
Learning Rate	0.01
Optimizer	AdamW

Dataset and Training

Dataset Statistics:

- 3,784 images collected
- 171,436 annotated objects
- 7 strategic locations
- 21 vehicle categories

Training Configuration:

- YOLOv11m model
- 256 epochs
- 640x640 resolution

Algorithm: Real-Time Emergency Vehicle Detection and Traffic Management

1. Start
2. Start analyzing real-time CCTV footage from all lanes.
3. For each lane, detect:
 - Number of Vehicles
 - Number of Emergencies Vehicles
 - Number of Pedestrians
4. If an emergency vehicle is detected:
 - Immediately open the corresponding lane.
 - If another emergency vehicle is detected in a different lane:
 - Wait until the first emergency lane is cleared.
5. For non-emergency scenarios:
 - Detect the number of vehicles and persons for each lane.
 - Calculate lane priority using the Weighted Job First algorithm.
 - Consider avoiding long starvation for any lane by ensuring:
 - Each lane stays open for a minimum time limit once opened.
 - No lane exceeds its maximum time limit.
6. Open the lane with the highest priority based on the calculated weights.
7. To manage traffic lights (green/yellow/red), send control signals to the hardware (e.g., Arduino, Raspberry Pi, NodeMCU).
8. Continue the process in a loop to maintain traffic signal control.
9. Stop

Model Performance Overview

Detection Accuracy:

- **Overall mAP50:** 79%
- **Vehicle Detection:** 85%
- **Pedestrian Detection:** 82%
- **Emergency Vehicle:** 85%

System Performance:

- **Response Time:** 222ms avg
- **System Uptime:** 98.9%
- **Processing Rate:** 30+ FPS
- **Memory Usage:** 2.1GB

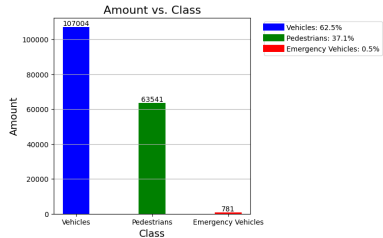
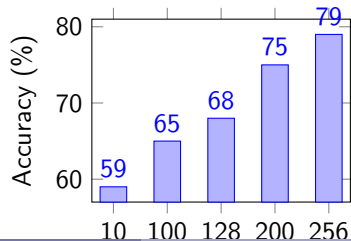


Figure: Performance Metrics



Confusion Matrix Analysis

Table: Model Confusion Matrix

True/Pred	Emerg	Person	Vehicle	BG
Emergency	4	3	0	1
Person	0	2080	47	890
Vehicle	0	41	4111	1191
Background	0	1152	0	1222

Performance Metrics:

- **Precision:** 83.2%
- **Recall:** 81.7%
- **F1-Score:** 82.4%
- **Training Time:** 45.6 hours

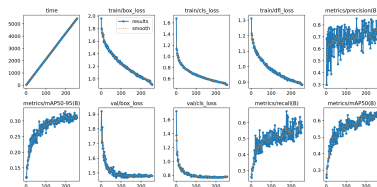


Figure: Confusion Matrix Visualization

Performance Improvements

Table: Traffic Flow Improvements

Metric	Traditional	Proposed	Improvement
Wait Time	12.5 min	8.4 min	33% ↓
Emergency	8.5 min	3.7 min	56% ↓
Lane Util.	70%	92%	31% ↑
Starvation	3.2/hr	0.1/hr	97% ↓

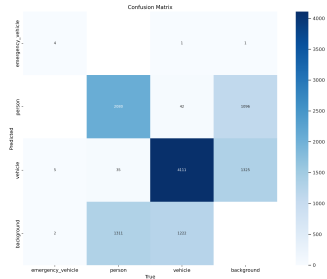


Figure: Performance Comparison

Significant improvements in all key metrics

Real-World Deployment Results

Pilot Locations:

1. Shahbag Intersection
2. Motijheel Commercial Area
3. Science Lab Junction

Deployment Duration:

- 2-month pilot study
- 24/7 continuous operation
- Multiple weather conditions
- Peak & off-peak hours

User Feedback:

- 87% overall satisfaction
- 92% noticed traffic improvements
- 79% support system expansion

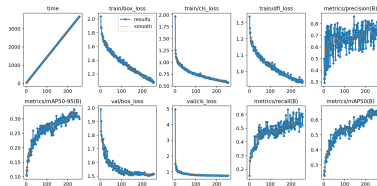


Figure: Deployment Results

System Reliability:

Location	Uptime	Response
Shahbag	98.7%	245ms
Motijheel	99.2%	198ms
Science Lab	98.9%	223ms
Average	98.9%	222ms

Technical Challenges & Solutions

Data Challenges:

- **Challenge:** Class imbalance (0.5% emergency vehicles)
- **Solution:** Data augmentation & weighted training

Hardware Limitations:

- **Challenge:** Processing power constraints
- **Solution:** Model optimization & edge computing

Environmental Factors:

- **Challenge:** Weather & lighting variations
- **Solution:** Robust training dataset & preprocessing

Integration Issues:

- **Challenge:** Legacy traffic infrastructure
- **Solution:** Modular IoT architecture

Real-time Requirements:

- **Challenge:** Sub-second response needed
- **Solution:** Optimized algorithms & parallel processing

Scalability Concerns:

- **Challenge:** City-wide deployment costs
- **Solution:** Cost-effective hardware & phased rollout

Cost-Benefit Analysis

- **Annual Net Benefits:** \$8.065 million
- **Return on Investment:** 672%
- **CO2 Reduction:** 1,245 tons annually
- **Fuel Savings:** 485,000 liters annually
- **User Satisfaction:** 87% overall satisfaction

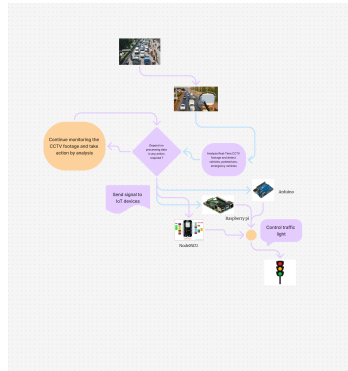
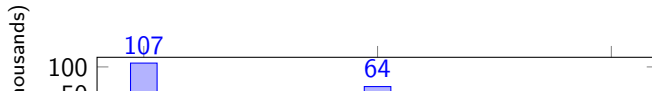


Figure: Economic Impact Analysis



Key Contributions

1. Novel YOLOv11 Application

- First comprehensive application for Dhaka traffic
- 79% detection accuracy achieved

2. Emergency Vehicle Prioritization

- 56% reduction in response times
- Automated detection and control

3. Adaptive Traffic Management

- 33% reduction in wait times
- 97% reduction in lane starvation

4. Cost-Effective Solution

- 672% return on investment
- Commercial hardware components

Future Research Directions

Short-term (6-12 months):

- Emergency vehicle dataset expansion
- Weather-resistant algorithms
- Hardware optimization
- User interface improvements
- Security enhancements

Medium-term (1-2 years):

- Predictive traffic modeling
- V2I communication integration
- Advanced ML algorithms
- Mobile app development
- Real-time analytics dashboard

Long-term (2-5 years):

- City-wide deployment
- Smart city ecosystem integration
- Autonomous vehicle compatibility
- International standardization
- AI-driven traffic optimization

Research Opportunities:

- Federated learning implementation
- Edge computing optimization
- Behavioral pattern analysis
- Environmental impact modeling
- Policy framework development

Questions & Discussion

Thank You!

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